Case Study 3: Detecting Generator Usage from Sensor Data.

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Background

Many African health facilities depend on different electricity sources

Reliable tracking of which **source** is **active** is essential for:

- **Monitoring fuel usage and carbon emissions**
- **Evaluating solarization impact**
- **Supporting energy planning**

Challenge

We need a reliable, scalable way to identify the active power source at any given time.

Current Approach: Rule-Based Detection

A rule-based algorithm classifies each reading as **Grid**, **Solar**, or **Generator** based on:

- Frequency thresholds
- Frequency deviation from grid reference (≤0.11 Hz)
- Frequency variability (MAD ≤ 0.1 Hz)
- Known room power source configuration

Existing solar PV system at Kabala Government Hospital, an off-grid health facility (credit: Albert Moiwa).

Rule Based Algorithm at a Glance

Strengths

- Simple, interpretable, and transparent
- Tuned for high accuracy within Sierra Leone
- Relies on expert domain knowledge

Limitations

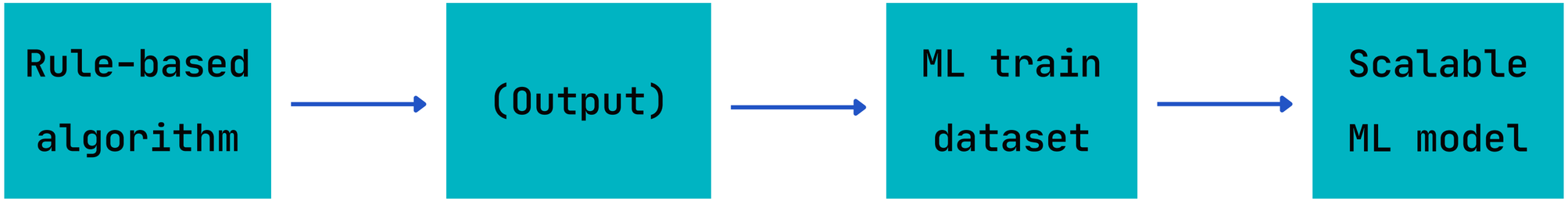
- Hard-coded thresholds → not transferable to other contexts and regions
- Frequency-only dependence → ignores other signals like voltage
- Manual smoothing → not scalable

A Generalizable Machine Learning Model

To expand to new hospitals and countries, we need a model that can:

- **Learn patterns automatically**
- **Adapt to different grids and generator behavior**
- **Maintain or improve accuracy without manual reconfiguration**

The Proposed Machine Learning Approach



<u>Process Overview</u>

1. Understanding and cleaning data.
2. Exploratory Data Analysis (EDA).
3. Feature engineering and development
4. Machine learning model.

Available Data Overview

Data collected every 2 minutes from 12 hospitals (57 unique rooms).

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Power Source configurations

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Available Data Overview

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Power Source configurations

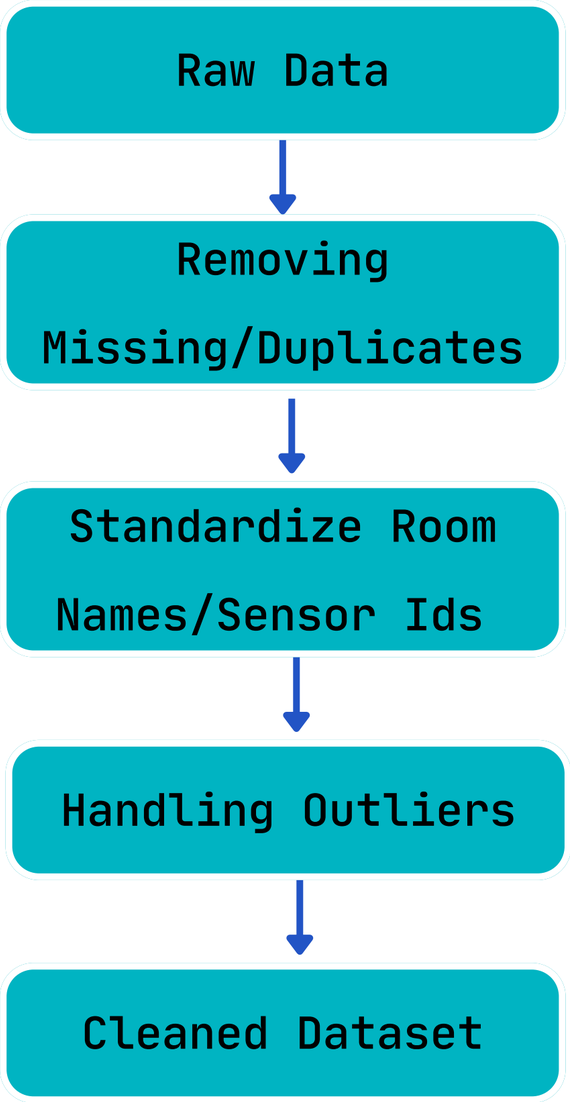
8 Unique features

7.9 million rows of data with 8 columns

Timeline: May 2023 - April 2024

Data Cleaning & Preparation

We cleaned over 7.9M sensor readings from 17 hospitals, ensuring a consistent, reliable dataset for feature engineering and model training.

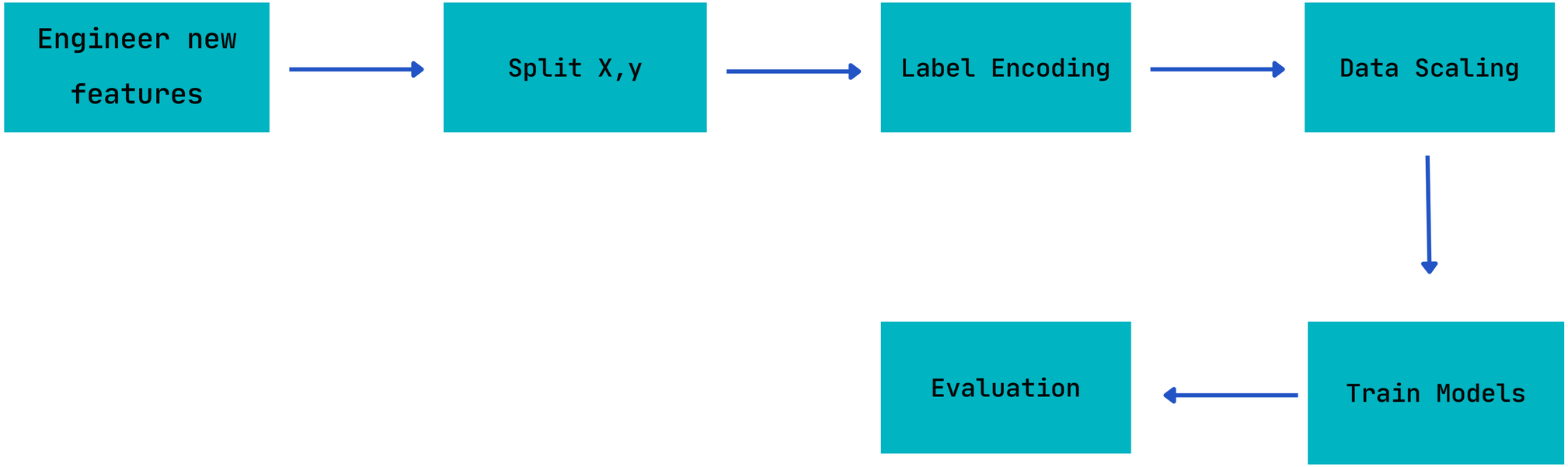


Feature engineering

Goal: Add features that reveal power source patterns

Feature name	Description	Purpose
Frequency Deviation from Grid Median	Absolute difference between each reading and the grid-only median (≈ 50.16 Hz)	Identifies how far a reading deviates from normal grid behavior.
Mean Absolute Deviation (MAD)	Variability of frequency within a 30-minute sliding window	Measures short-term stability (solar tends to have low variability, generators have high)
Power Source Configuration	Known setup of each room (eg solar /Grid Backup)	Adds room setup informartion that helps the model tell apart cases with more than one power sources

Model Training with Enhanced Features



Classification Models

Logistic Regression	Simple baseline model for comparison with more complex approaches.
Decision Tree Classifier	Chosen for its ability to handle non-linear relationships and provide interpretable decision rules
Random Forest Classifier	An ensemble of trees that improves accuracy, reduces overfitting, and performs well with large datasets
K-Nearest Neighbors	Captures local similarity patterns in the data, making it useful for structured numeric features like voltage and frequency.

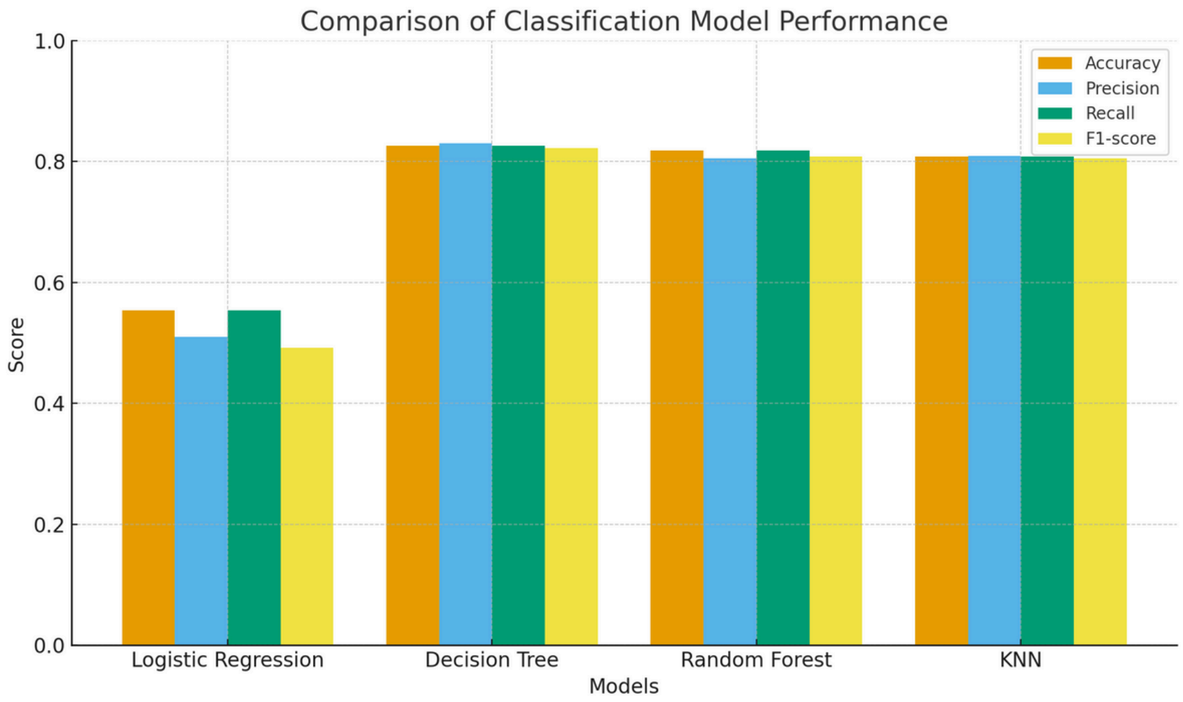
Perfomance Metrics

Accuracy	Measures overall model correctness across all power sources
Precision	Ensures the model’s predictions are trustworthy with few false positives
Recall	Confirms the model captures most true cases of each power source
F1-Score	Balances precision and recall for consistent performance

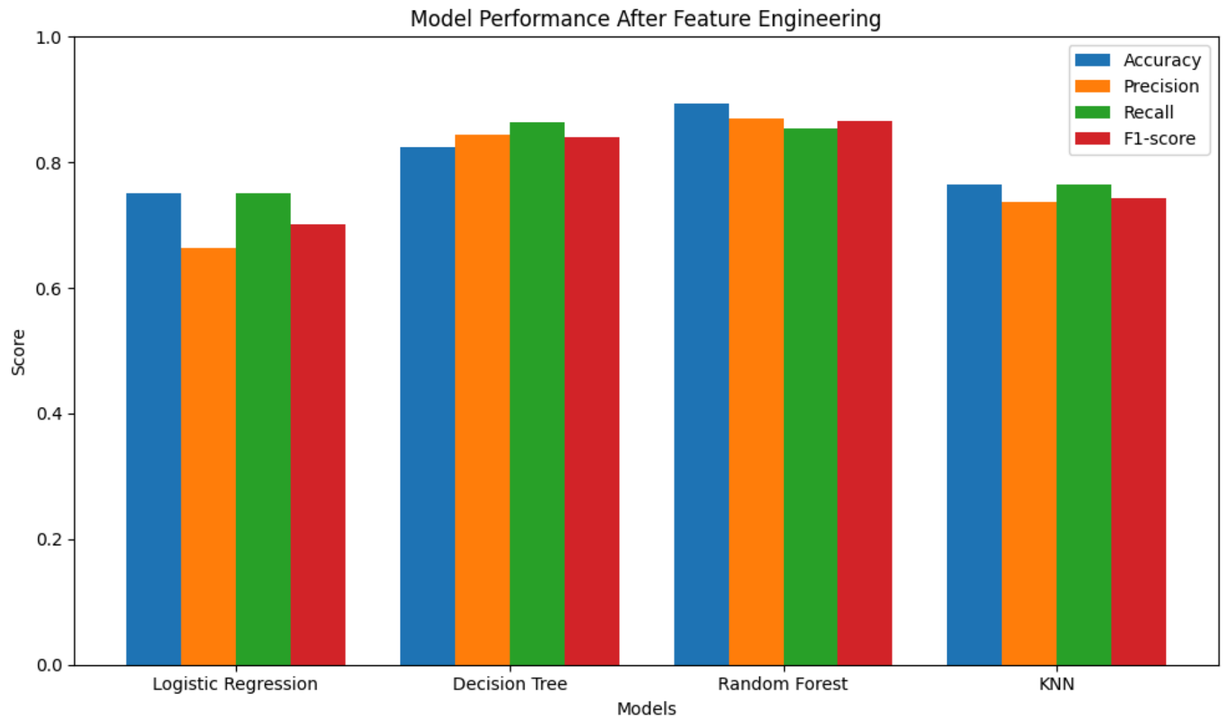
Accuracy and **F1-score** were chosen because the data is balanced and they together show overall performance and how well each power source is correctly detected.

Performance Metrics Comparison

Before Feature Engineering



After Feature Engineering



Recommendation

The **Random Forest model** achieved an **accuracy** of **89.98%** and an **F1-score** of **89.91%**, showing strong and consistent performance. This indicates that the model **effectively learns patterns** from the data and engineered features, making it reliable for power source detection and **scalable** to **new hospitals and regions.**

Solution Impact

Determining the active power source is crucial because it enables the computation of solar, grid, and generator usage time.

- Impact of solarization projects
- Reliable monitoring of energy sources
- Better hospital planning & budgeting
- Estimating power consumption of different power sources

Thank You